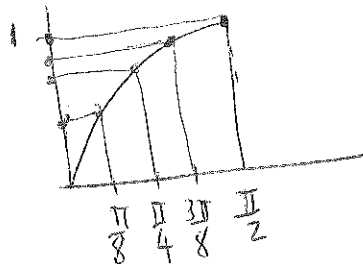


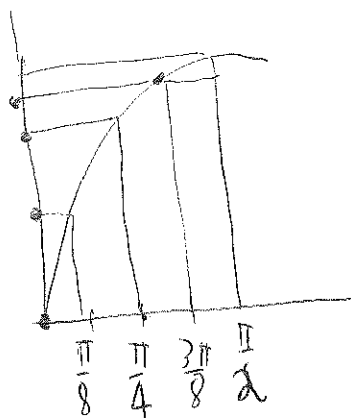
Unit 5.2

4 a) right endpoints



$$R_4 = \frac{\pi}{8} \left[\sin \frac{\pi}{8} + \sin \frac{\pi}{4} + \sin \frac{3\pi}{8} + \sin \frac{4\pi}{8} \right]$$

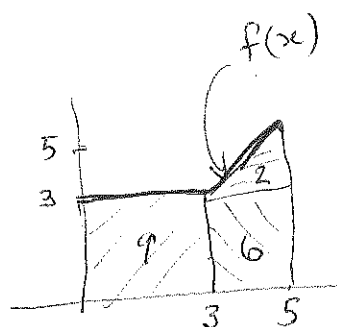
b)



$$L_4 = \frac{\pi}{8} \left[\sin 0 + \sin \frac{\pi}{8} + \sin \frac{\pi}{4} + \sin \frac{3\pi}{8} \right]$$

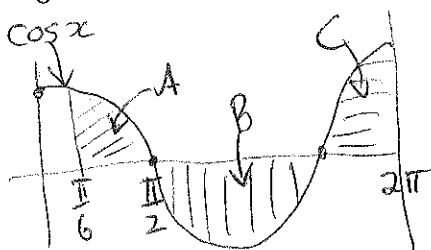
Unit 5.3 Ex 5.2

$$f(x) = \begin{cases} 3 & \text{for } x < 3 \\ x & \text{for } x \geq 3 \end{cases}$$



$$\int_0^5 f(x) = \text{area in sketch} = 9 + 6 + 2 = 17$$

Unit 5.4 Ex 5.3 no 54



$$\int_{\pi/6}^{2\pi} \cos x dx = \sin x \Big|_{\pi/6}^{2\pi} = \sin 2\pi - \sin \frac{\pi}{6} = 0 - \frac{1}{2} = -\frac{1}{2}$$

with the sketch

$$\int_{\pi/6}^{2\pi} \cos x dx = \text{area A} + \text{area C} - \text{area B}$$